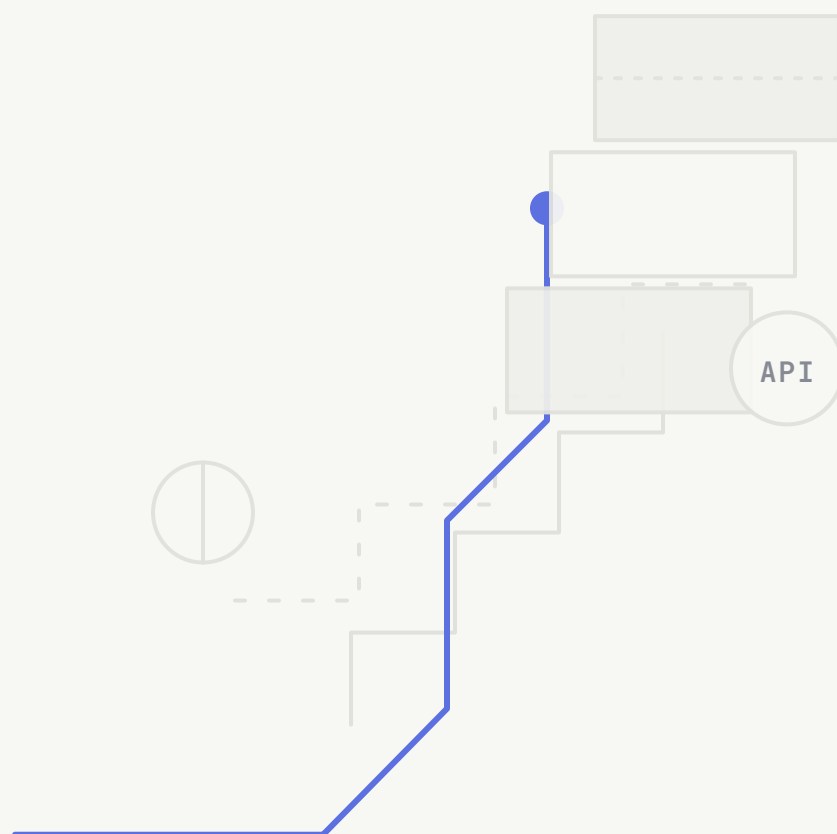


RAG is a data system.

If retrieval is wrong, a smarter model only explains the mistake more fluently.

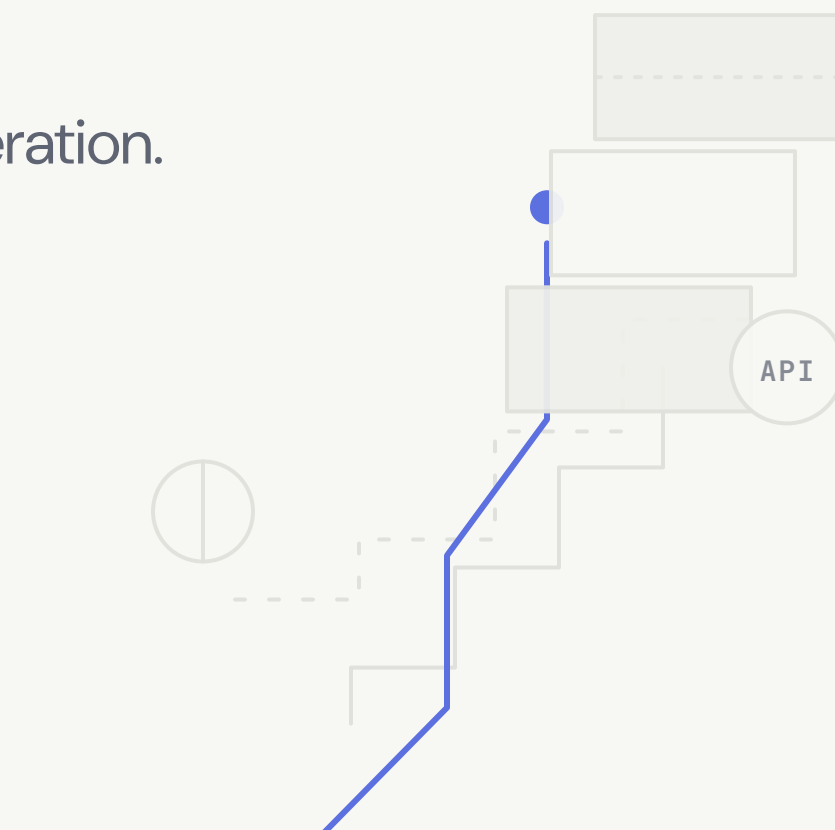
SWIPE →



01

Measure retrieval before generation.

A good answer can hide a bad top-k.
Log the retrieved chunks before generation.



Freshness is a retrieval feature.

CONCEPT

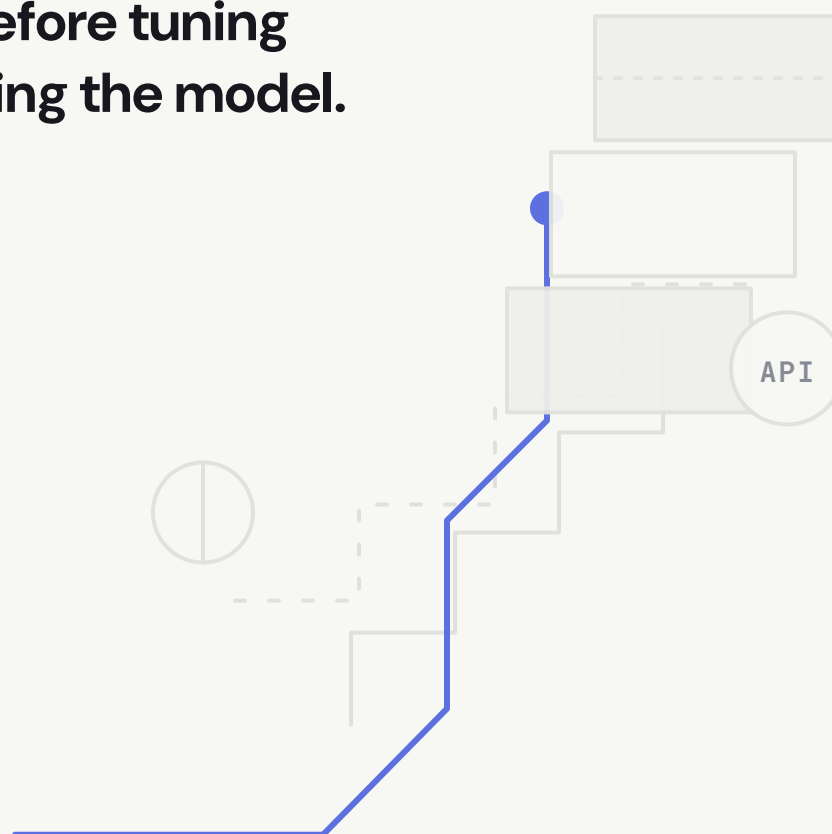
An index is part of the product.
Stale documents become stale answers.

MECHANISM

Track document age and index version.
Refresh on a known schedule.

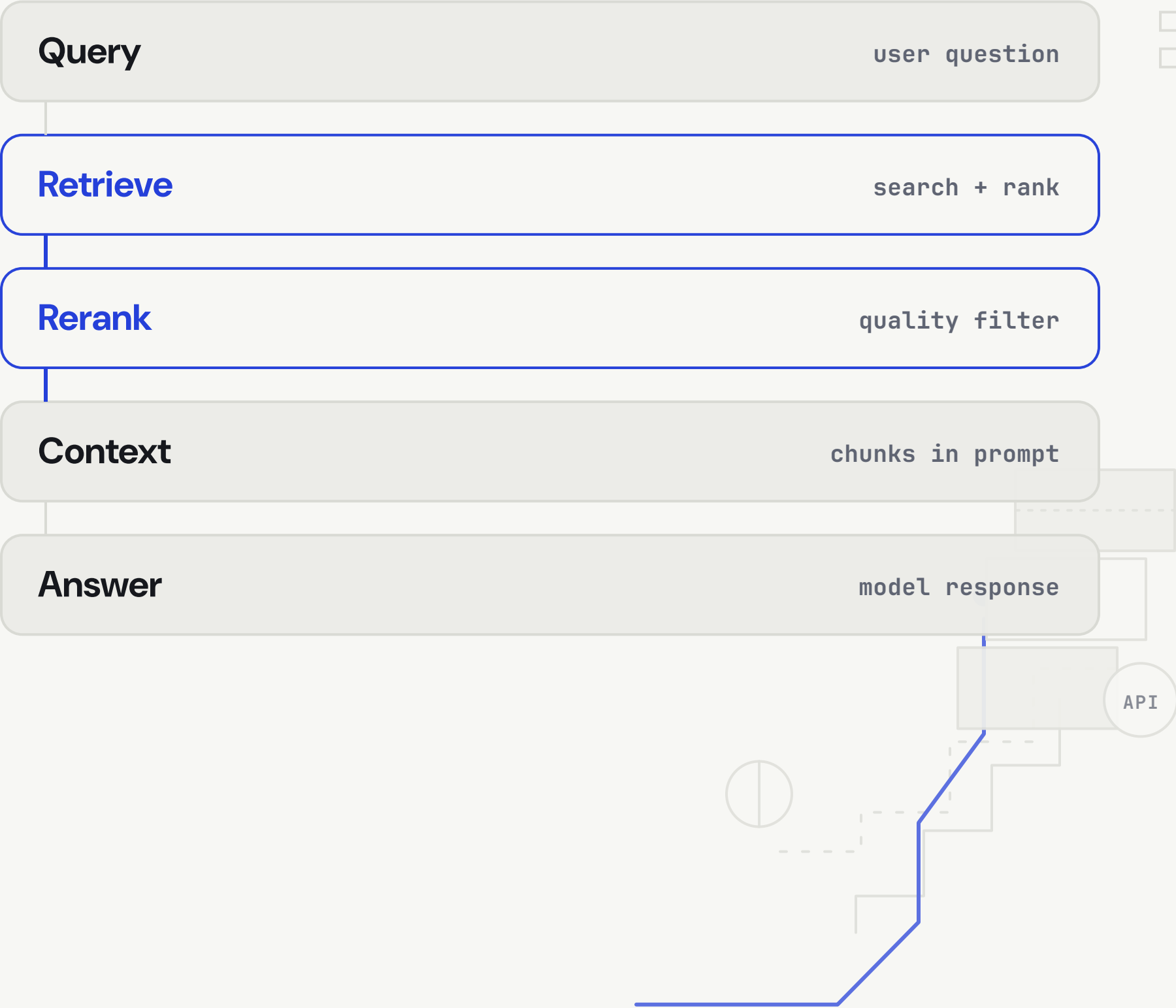
TAKEAWAY

**Measure freshness before tuning
the prompt or swapping the model.**



Where a RAG answer can fail

The blue path is the first place I inspect.



RAG vs fine-tuning

RAG

Knowledge changes often

Need fresh citations

Gap is behavior

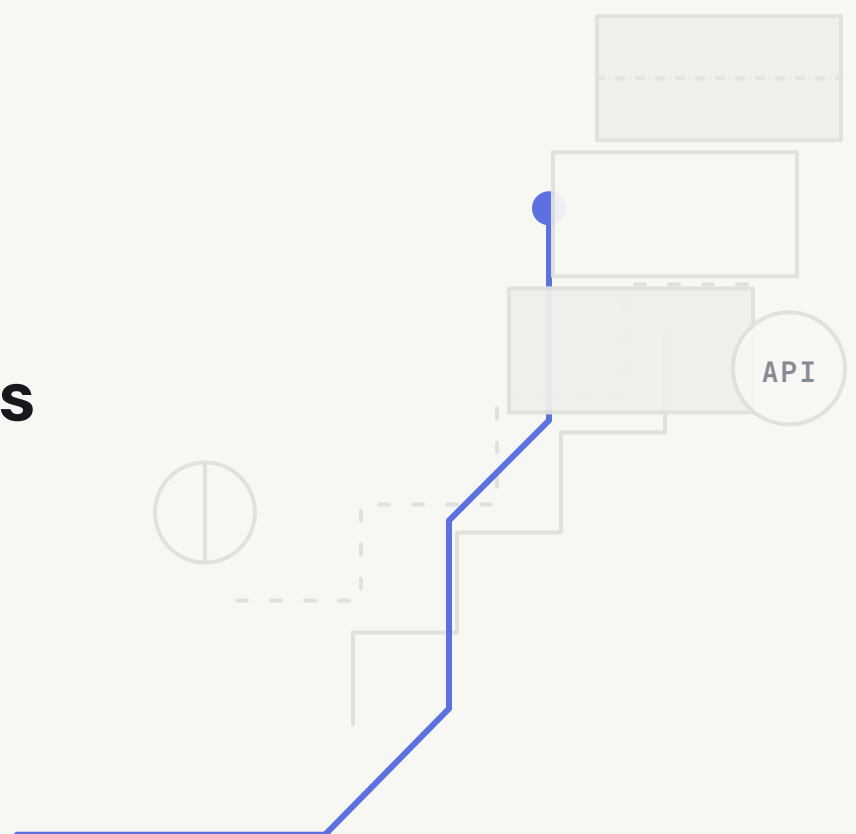
FINE-TUNING

Output format must be stable

Per-call cost matters

Gap is knowledge

Start with RAG when the gap is knowledge, not behavior.



Fix retrieval first. Then tune the prompt.

A traceable retrieval path gives you
a better debugging surface.

Save this for your next RAG review.

M. Ahsan Izhar

Software Engineering × AI Integration